

## 2022 AIME I Problems/Problem 4

### Contents

- [1 Problem](#)
- [2 Solution 1](#)
- [3 Solution 2](#)
- [4 Video Solution](#)
- [5 Video Solution \(Mathematical Dexterity\)](#)
- [6 Video Solution](#)
- [7 Video Solution](#)
- [8 Video Solution by MRENTHUSIASM \(English & Chinese\)](#)
- [9 Video Solution](#)
- [10 See Also](#)

### Problem

Let  $w = \frac{\sqrt{3} + i}{2}$  and  $z = \frac{-1 + i\sqrt{3}}{2}$ , where  $i = \sqrt{-1}$ . Find the number of ordered pairs  $(r, s)$  of positive integers not exceeding 100 that satisfy the equation  $i \cdot w^r = z^s$ .

### Solution 1

We rewrite  $w$  and  $z$  in polar form:

$$w = e^{i \cdot \frac{\pi}{6}},$$

$$z = e^{i \cdot \frac{2\pi}{3}}.$$

The equation  $i \cdot w^r = z^s$  becomes

$$e^{i \cdot \frac{\pi}{2}} \cdot \left(e^{i \cdot \frac{\pi}{6}}\right)^r = \left(e^{i \cdot \frac{2\pi}{3}}\right)^s$$

$$e^{i\left(\frac{\pi}{2} + \frac{\pi}{6}r\right)} = e^{i\left(\frac{2\pi}{3}s\right)}$$

$$\frac{\pi}{2} + \frac{\pi}{6}r = \frac{2\pi}{3}s + 2\pi k$$

$$3 + r = 4s + 12k$$

$$3 + r = 4(s + 3k).$$

for some integer  $k$ .

Since  $4 \leq 3 + r \leq 103$  and  $4 \mid 3 + r$ , we conclude that

$$3 + r \in \{4, 8, 12, \dots, 100\},$$

$$s + 3k \in \{1, 2, 3, \dots, 25\}.$$

Note that the values for  $s + 3k$  and the values for  $r$  have one-to-one correspondence.

We apply casework to the values for  $s + 3k$  :

$$1. s + 3k \equiv 0 \pmod{3}$$

There are 8 values for  $s + 3k$ , so there are 8 values for  $r$ . It follows that  $s \equiv 0 \pmod{3}$ , so there are 33 values for  $s$ .

There are  $8 \cdot 33 = 264$  ordered pairs  $(r, s)$  in this case.

$$2. s + 3k \equiv 1 \pmod{3}$$

There are 9 values for  $s + 3k$ , so there are 9 values for  $r$ . It follows that  $s \equiv 1 \pmod{3}$ , so there are 34 values for  $s$ .

There are  $9 \cdot 34 = 306$  ordered pairs  $(r, s)$  in this case.

$$3. s + 3k \equiv 2 \pmod{3}$$

There are 8 values for  $s + 3k$ , so there are 8 values for  $r$ . It follows that  $s \equiv 2 \pmod{3}$ , so there are 33 values for  $s$ .

There are  $8 \cdot 33 = 264$  ordered pairs  $(r, s)$  in this case.

Together, the answer is  $264 + 306 + 264 = \boxed{834}$ .

~MRENTHUSIASM

## Solution 2

First we recognize that  $w = \text{cis}(30^\circ)$  and  $z = \text{cis}(120^\circ)$  because the cosine and sine sums of those angles give the values of  $w$  and  $z$ , respectively. By De Moivre's theorem,  $\text{cis}(\theta)^n = \text{cis}(n\theta)$ . When you multiply by  $i$ , we can think of that as rotating the complex number  $90^\circ$  counterclockwise in the complex plane. Therefore, by the equation we know that  $30r + 90$  and  $120s$  land on the same angle.

This means that

$$30r + 90 \equiv 120s \pmod{360},$$

which we can simplify to

$$r + 3 \equiv 4s \pmod{12}.$$

Notice that this means that  $r$  cycles by 12 for every value of  $s$ . This is because once  $r$  hits 12, we get an angle of  $360^\circ$  and the angle laps onto itself again. By a similar reasoning,  $s$  laps itself every 3 times, which is much easier to count. By listing the possible values out, we get the pairs  $(r, s)$ :

$$\begin{array}{cccccccc} (1, 1) & (5, 2) & (9, 3) & (13, 1) & (17, 2) & (21, 3) & \dots & (97, 1) \\ (1, 4) & (5, 5) & (9, 6) & (13, 4) & (17, 5) & (21, 6) & \dots & (97, 4) \\ (1, 7) & (5, 8) & (9, 9) & (13, 7) & (17, 8) & (21, 9) & \dots & (97, 7) \\ \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots & \vdots \\ (1, 100) & (5, 98) & (9, 99) & (13, 100) & (17, 98) & (21, 99) & \dots & (97, 100) \end{array}$$

We have 25 columns in total: 34 values for the first column, 33 for the second, 33 for the third, and then 34 for the fourth, 33 for the fifth, 33 for the sixth, etc. Therefore, this cycle repeats every 3 columns and our total sum is

$$(34 + 33 + 33) \cdot 8 + 34 = 100 \cdot 8 + 34 = \boxed{834}.$$

~KingRavi

## Video Solution

[2022 AIME I #4](#)

[MathProblemSolvingSkills.com](https://www.mathproblemsolving.com)

## Video Solution (Mathematical Dexterity)

<https://www.youtube.com/watch?v=XiEaCq5jf5s>

## Video Solution

<https://www.youtube.com/watch?v=qQ0TIhHuhnl>

~Steven Chen ([www.professorchen.edu](http://www.professorchen.edu))

## Video Solution

[https://youtu.be/MJ\\_M-xvwHLk?t=933](https://youtu.be/MJ_M-xvwHLk?t=933)

~ThePuzzlr

## Video Solution by MRENTHUSIASM (English & Chinese)

[https://www.youtube.com/watch?v=1Z6GbkbFu4Q&ab\\_channel=MRENTHUSIASM](https://www.youtube.com/watch?v=1Z6GbkbFu4Q&ab_channel=MRENTHUSIASM)

~MRENTHUSIASM

## Video Solution

[https://youtu.be/m1vg\\_DfHEX4](https://youtu.be/m1vg_DfHEX4)

~AMC & AIME Training

## See Also

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Preceded by  
[Problem 3](#)

Followed by  
[Problem 5](#)

[1](#) • [2](#) • [3](#) • [4](#) • [5](#) • [6](#) • [7](#) • [8](#) • [9](#) • [10](#) • [11](#) • [12](#) • [13](#) • [14](#) • [15](#)

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